# Evolve the assets, solve the weights

A hybrid metaheuristic and quadratic programming approach to large-universe portfolio selection

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Replicates and extends Hidalgo-Marín, Nebro & García-Nieto, *Efficient Selection of Investment Portfolios in Real-World Markets*, Algorithms 2026, 19, 20 [1]

## 1 The problem

Choose $k$ of $N$ assets and their weights to minimise risk and maximise return:

$$\min\,(\sigma, -R), \quad \sigma = \sqrt{250\, w^{\top}\Sigma_S\, w}, \quad R = \textstyle\sum_{j\in S} w_j\mu_j$$

$$|S| = k, \qquad \textstyle\sum_j w_j = 1, \qquad \frac{1}{2k} \le w_j \le \frac{2}{k}$$

The paper's 12 instances use 2023 daily prices of U.S.-listed assets. The largest, **ALL_20_6000**, picks $k = 20$ of $N = 6000$.

### The original study [1]

- NSGA-II, MOEA/D, SMS-EMOA and SMPSO (jMetal), 30 runs each.
- Real-valued genes: $k$ pointers into the return-sorted asset list and $k$ weights.
- SMS-EMOA ranks best. On ALL_20_6000 only SMPSO reaches the high-return end.

### Our replica (pymoo)

- The three evolutionary algorithms with the paper's encoding and settings, 10 runs each.
- Same ranking (SMS-EMOA, NSGA-II, MOEA/D) and practically the same fronts: hypervolume 0.944 for both on ALL_20_6000. Gains below come from the changes, not the framework.

## 2 What we changed

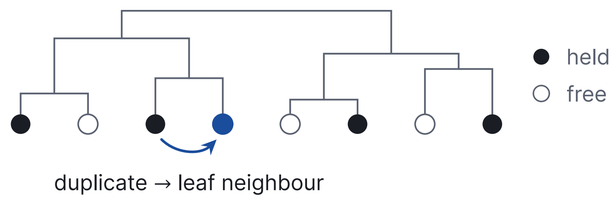
| | Original paper [1] | This work |
|---|---|---|
| **Encoding** | $2k$ reals: pointers $s_i \in [0,1]$ pick asset $\lfloor s_i N \rfloor$ of the return-sorted list, plus $k$ weights | **Set-based**: $k$ distinct asset IDs, plus one trade-off gene $t \in [0,1]$ |
| **Crossover** | SBX on all genes ($p_c = 0.9$, $\eta = 20$) | **Uniform slot crossover**: each asset comes from either parent |
| **Mutation** | Polynomial ($p_m = 1/n$, $\eta = 20$) | **Swap**: a held asset is exchanged for one outside the portfolio |
| **Duplicates** | Next index of the return-sorted list | **HRP repair**: next free leaf of the correlation dendrogram |
| **Weights** | Evolved, then normalised and clipped | **Solved exactly** by a QP for the chosen assets |
| Same | algorithms, settings, evaluation budgets (50 000–250 000 per run), seeds, objectives, constraints | |

### HRP collision repair

Slot crossover can give a child the same asset twice. As in Hierarchical Risk Parity [2], assets are clustered (single linkage) on

$$d_{ij} = \sqrt{\tfrac{1}{2}(1 - \rho_{ij})}$$

The leaf order puts correlated assets side by side, so a duplicate moves to a close substitute.

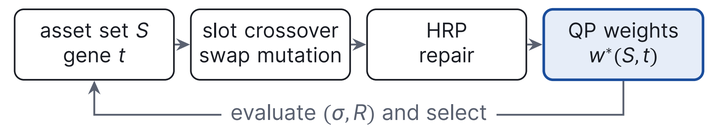



### Exact weights: splitting the MIQP

The metaheuristic picks the integer part (which assets). What is left is a convex QP in the 20 weights, solved exactly for every portfolio (quadprog [3]):

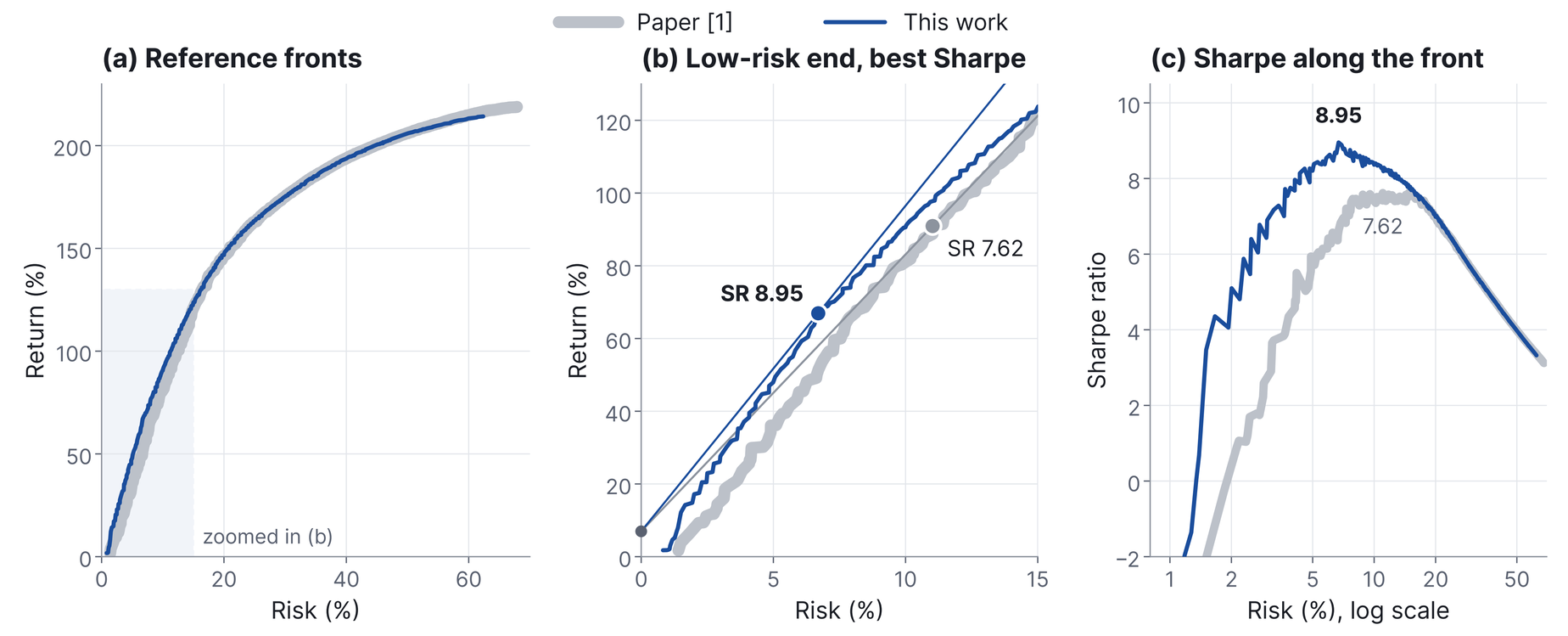
$$\min_{w}\ \gamma\,\sigma^2 - R, \quad \textstyle\sum_j w_j = 1,\ \frac{1}{2k} \le w_j \le \frac{2}{k}, \quad \gamma = 10^{-2+6t}$$

$t$ slides along the efficient frontier of the chosen assets.




## 3 Results on the largest instance: ALL_20_6000

6 000 assets, $k = 20$, 250 000 evaluations per run. A reference front merges all runs of all algorithms: the paper's 4 × 30 runs, this work's 3 × 10 runs.




**8.95**
**best Sharpe ratio**
paper 7.62 (+18 %)

**+12 pp**
**return at 5 % risk**
48 % vs 36 %

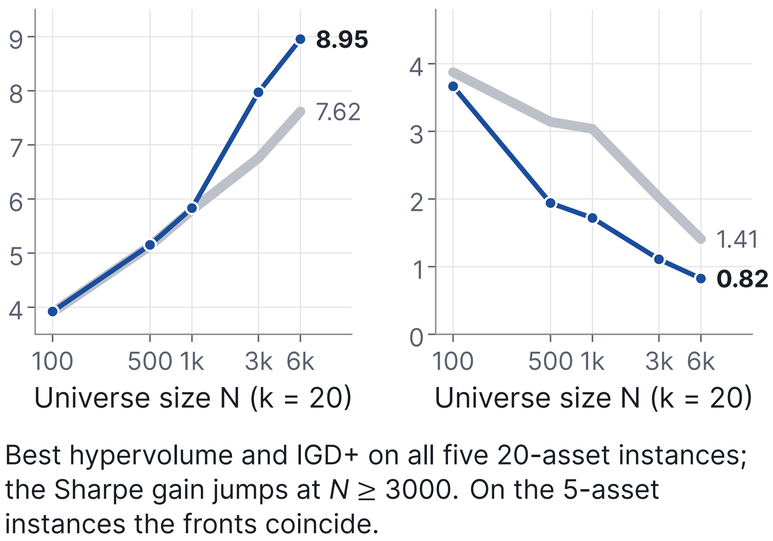
**0.82 %**
**lowest risk**
paper 1.41 %

**0.951**
**hypervolume**
paper 0.944

(b) The lines from the risk-free rate $R_f = 6.97\,\%$ touch each front at its best Sharpe ratio SR $= (R - R_f)/\sigma$, their slope. (c) The Sharpe ratio of every portfolio on each front.

The gain sits below 15 % risk, where 20 weights are hardest to evolve. Above it the fronts coincide; at the top this work reaches 214 % return, the paper 219 %.

## 4 Gains grow with $N$



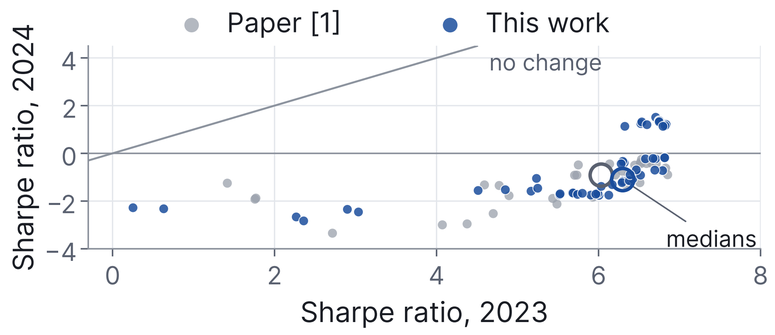

Best hypervolume and IGD+ on all five 20-asset instances; the Sharpe gain jumps at $N \ge 3000$. On the 5-asset instances the fronts coincide.

## 5 Robustness and cost

- **Per run**, all 12 instances: hypervolume significantly better than the replica's in 33 of 36 algorithm–instance pairs, never worse (Wilcoxon, 5 %). The paper publishes merged fronts only.
- **3 ties**, all on the largest universes (SMS-EMOA on ALL_20_3000 and ALL_20_6000, NSGA-II on ALL_20_6000). Single runs there reach lower risk but a shorter high-return end, since swap mutation samples 6 000 assets uniformly.
- **Cost**: same evaluation budget. The QPs raised the time for all 360 runs from 63 to 67 min.

## 6 Out of sample: 2024

Both ALL_20_6000 fronts on Yahoo Finance prices: portfolios with 80 % of their weight traded in 2023–24, leaving out those with a negative Sharpe ratio in both years.




Median Sharpe 2023 → 2024: 6.03 → −0.91 for the paper (32 of 658 portfolios), 6.30 → −1.10 for this work (54 of 369). The gain is in optimisation, not in forecasting.

## Conclusions

- A pymoo replica reproduces the paper's ranking and reference fronts.
- Evolving only the assets (set encoding, slot crossover, swap mutation, HRP repair) and solving the weights by QP gives the best front on every 20-asset instance, and the gain grows with $N$: on ALL_20_6000 the best Sharpe ratio rises from 7.62 to 8.95 at the same budget.
- The gain is in-sample: in 2024 both approaches lose most of their Sharpe ratio. Better estimates of returns and risk, rather than a better optimiser, are the next step.

## References

[1] A. J. Hidalgo-Marín, A. J. Nebro, J. García-Nieto. Efficient Selection of Investment Portfolios in Real-World Markets: A Multi-Objective Optimization Approach. *Algorithms* 2026, 19(1), 20. doi:10.3390/a19010020

[2] M. López de Prado. Building Diversified Portfolios that Outperform Out of Sample. *J. Portfolio Management* 2016, 42(4), 59–69.

[3] D. Goldfarb, A. Idnani. A numerically stable dual method for solving strictly convex quadratic programs. *Math. Programming* 1983, 27, 1–33.

[4] J. Blank, K. Deb. pymoo: Multi-Objective Optimization in Python. *IEEE Access* 2020, 8, 89497–89509.